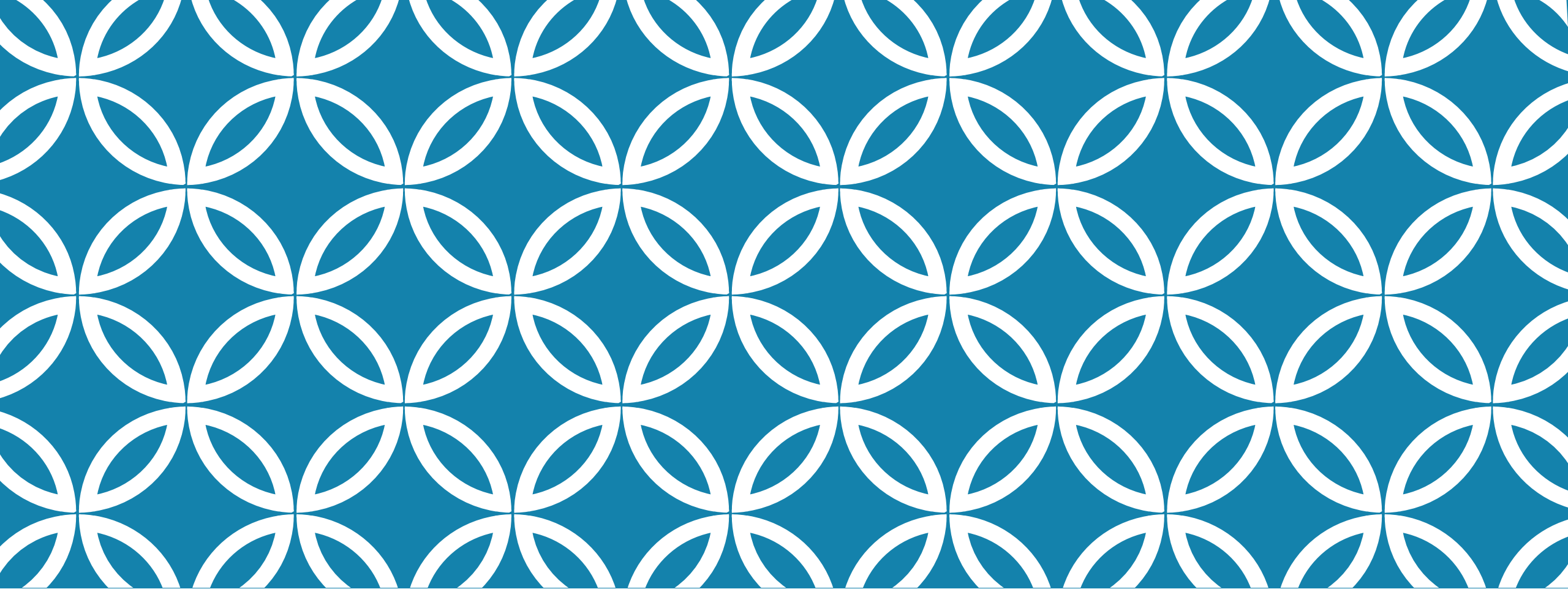




DATA ANALYTICS PROJECT

By Supoj Xu (Sean)

Airbnb Listing Trends and Pricing
Insights for First-Time Hosts

AGENDA

- Problem
- Goals
- Data
- Findings
- Recommendations



PROBLEM STATEMENT

First-time Airbnb hosts are unsure how and what they need to consider when pricing their rental on the platform.

GOALS

Goal 1: To extract and share relevant insights from the Airbnb listing data which can help with decision-making.

Goal 2: To provide pricing recommendations to first-time hosts so they can maximize their long-term rental revenue.

Main Questions:

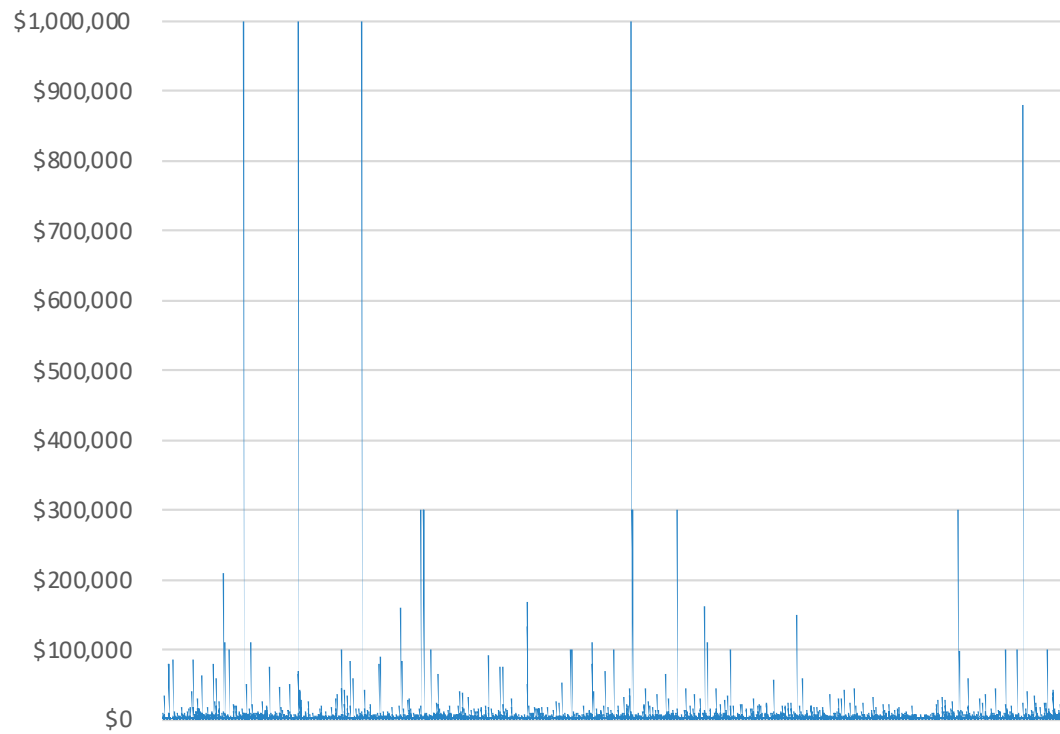
- What factors have more influence on price, and how do the relationships vary across different settings?
- How should first-time Airbnb host price their property so that they can get appropriate revenue?

DATA: OVERVIEW

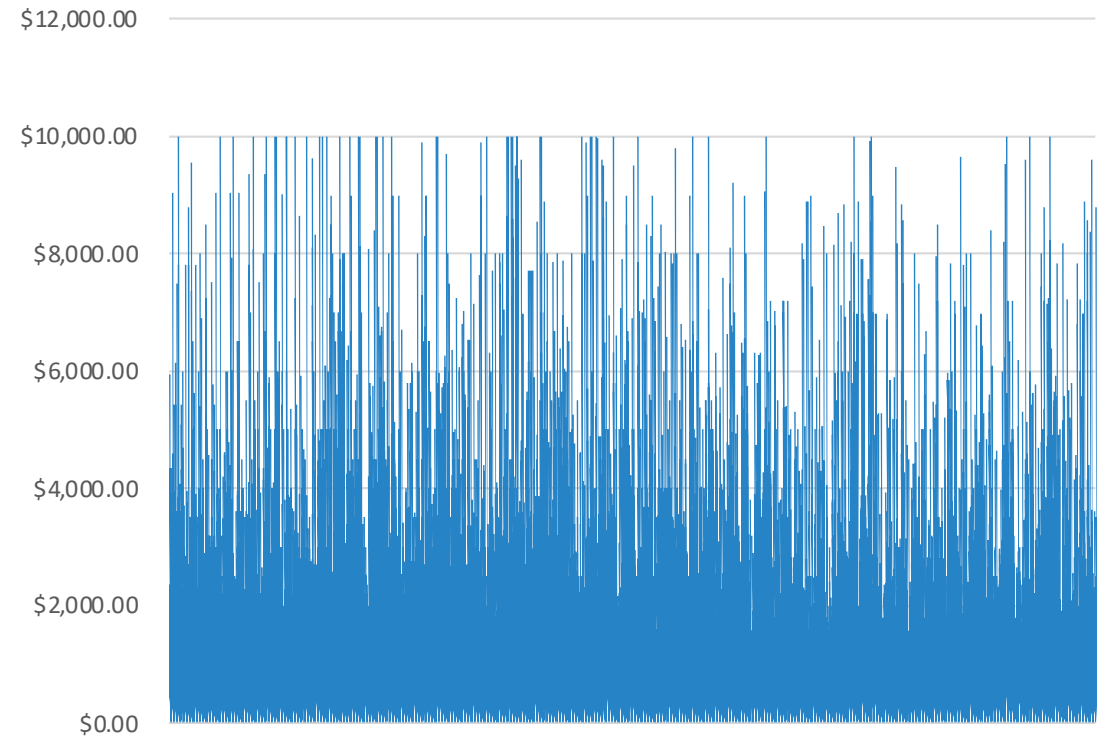
- **Data:** Airbnb Listings in Bangkok, Thailand
- **Source:** insideairbnb.com
- **Date:** September 22, 2023
- **Size:** 75 columns, 20,824 rows
- **Fields:** neighbourhood, room type, price, bedrooms, bathrooms, accommodates, minimum nights, number of reviews, etc.

DATA: EXTREME OUTLIERS

Price Distribution (Before)



Price Distribution (After)

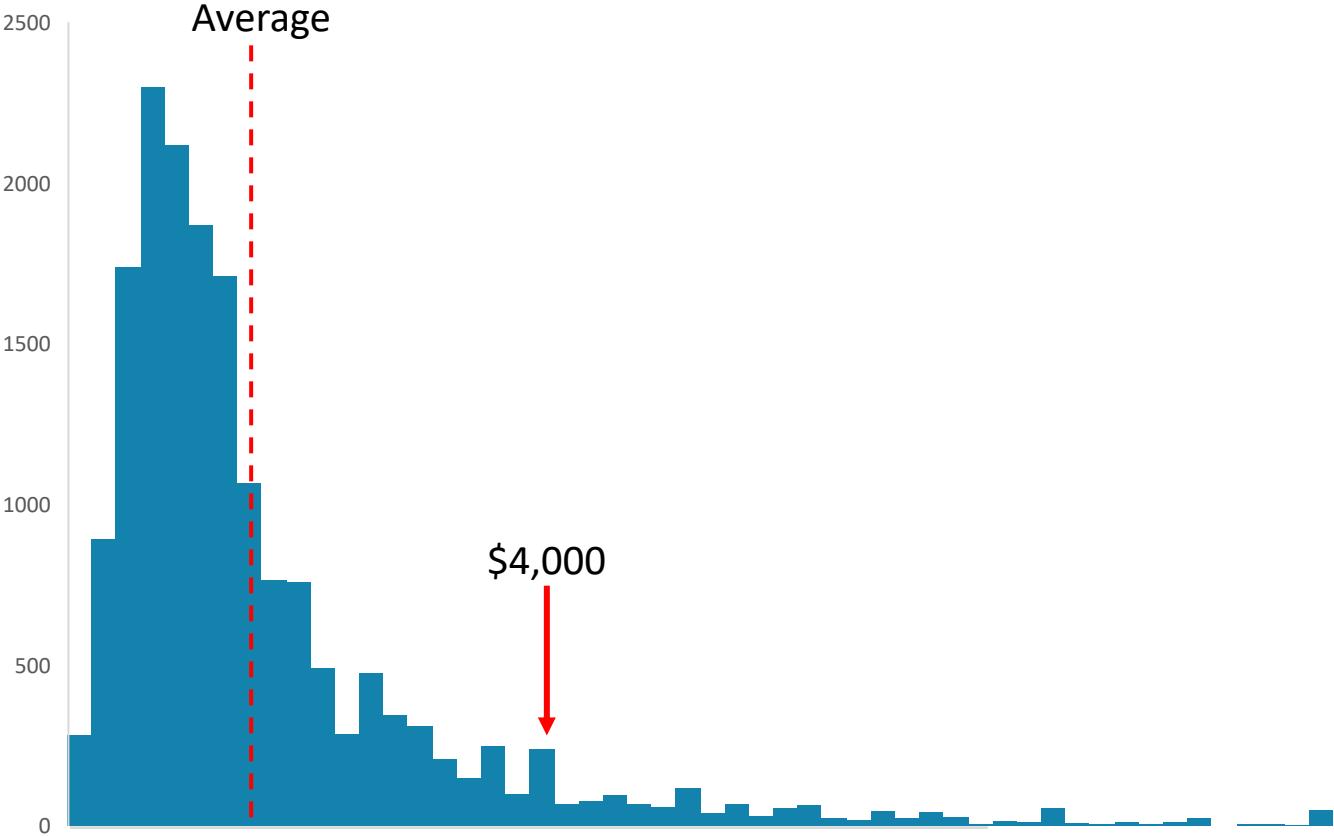


DATA: MAJOR CHANGES

- Dropped 48 irrelevant and incomplete columns
- **Removed all hotel-related listings (obvious ones)**
- Removed listings with certain values in an abnormal range
- Replaced values above 30 in “minimum_nights” with 30
- + Added “region” column to categorize districts into groups
- + Added “min_night_cat” to categorize minimum nights into groups

FINDINGS: PRICE OVERALL

Price Histogram

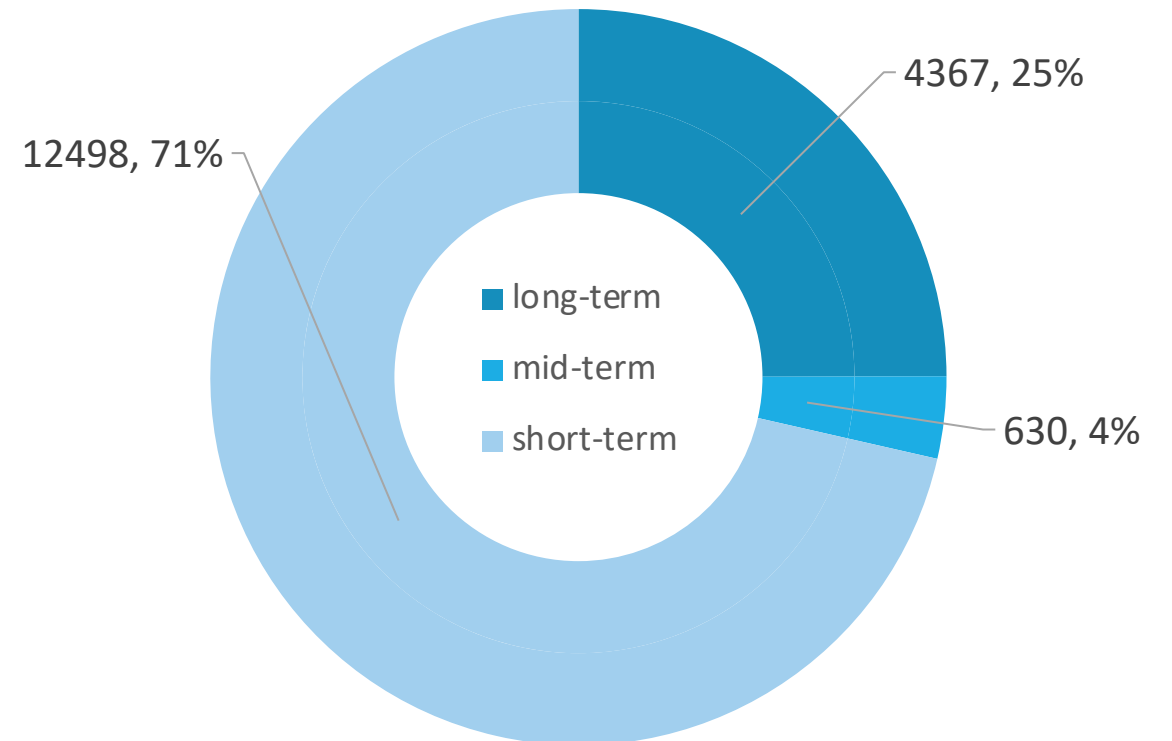


price (after cleaning)	
Mean	1748.05207
Standard Error	10.7841006
Median	1300
Mode	1000
Standard Deviation	1426.43937
Sample Variance	2034729.28
Kurtosis	8.35381215
Skewness	2.58078512
Range	10000
Minimum	216
Maximum	10000
Sum	30583919
Count	17496

FINDINGS: STAY TERM

Insight #1: Most listings are available for short-term stay

Listing Count by Stay Term



FINDINGS: STAY TERM

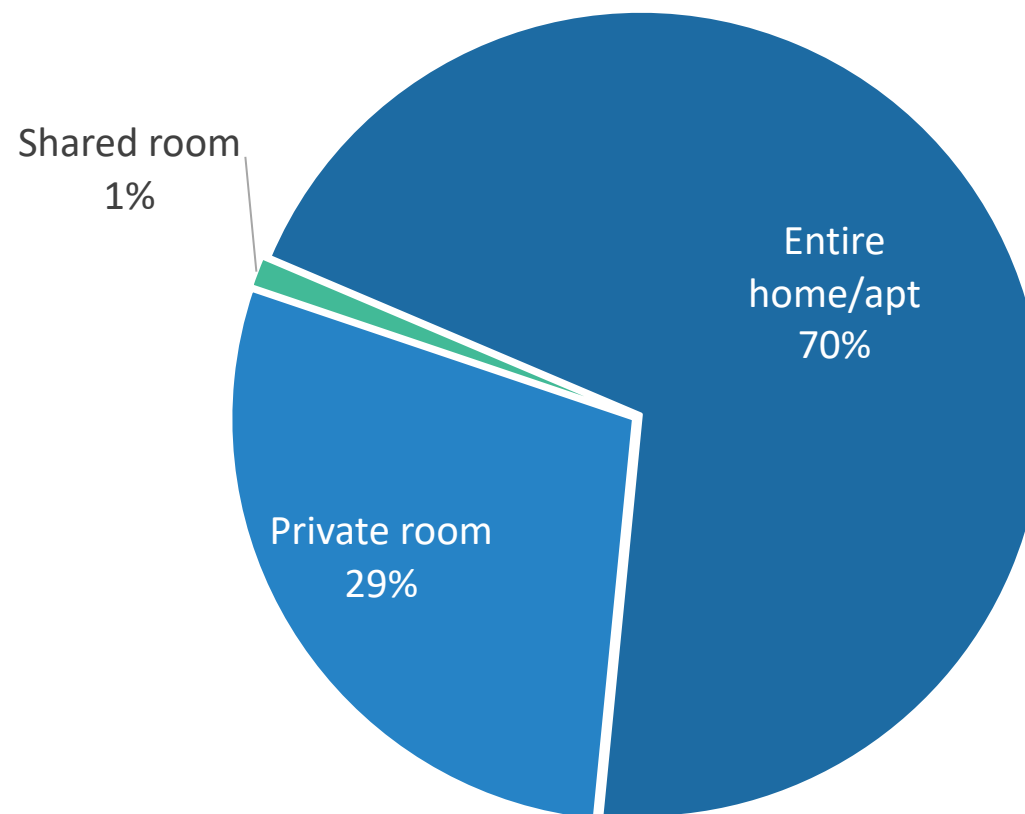
Insight #2: Short-term listing has the highest average price and average reviews per month.

Stay term	Average price	Reviews per month
long-term	1428.23	0.32
mid-term	1611.00	0.50
short-term	1866.85	0.80
total	1748.15	0.67

FINDINGS: ROOM TYPE

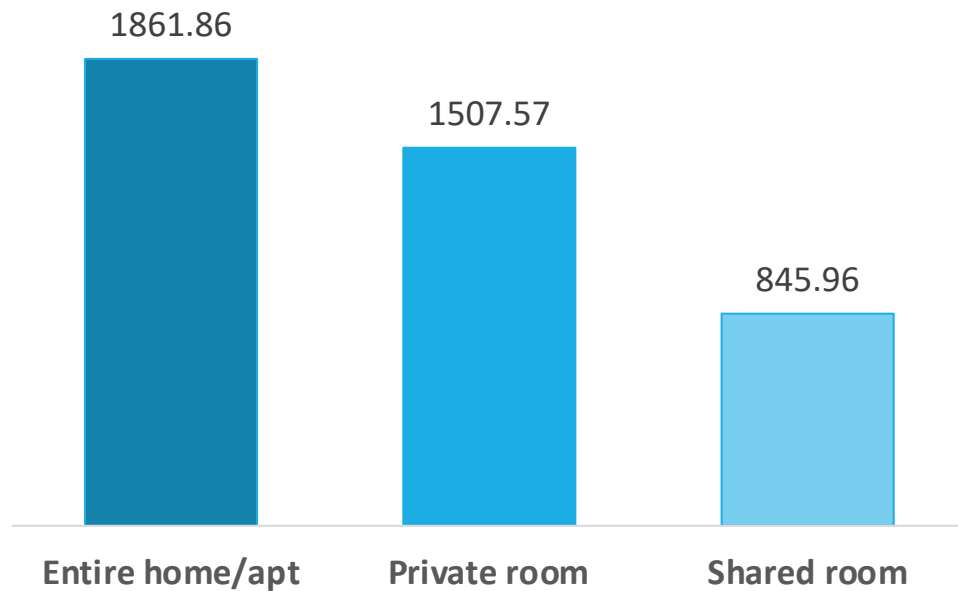
Insight #3: Most of the listings are entire home.

Listing Count by Room Type



FINDINGS: ROOM TYPE

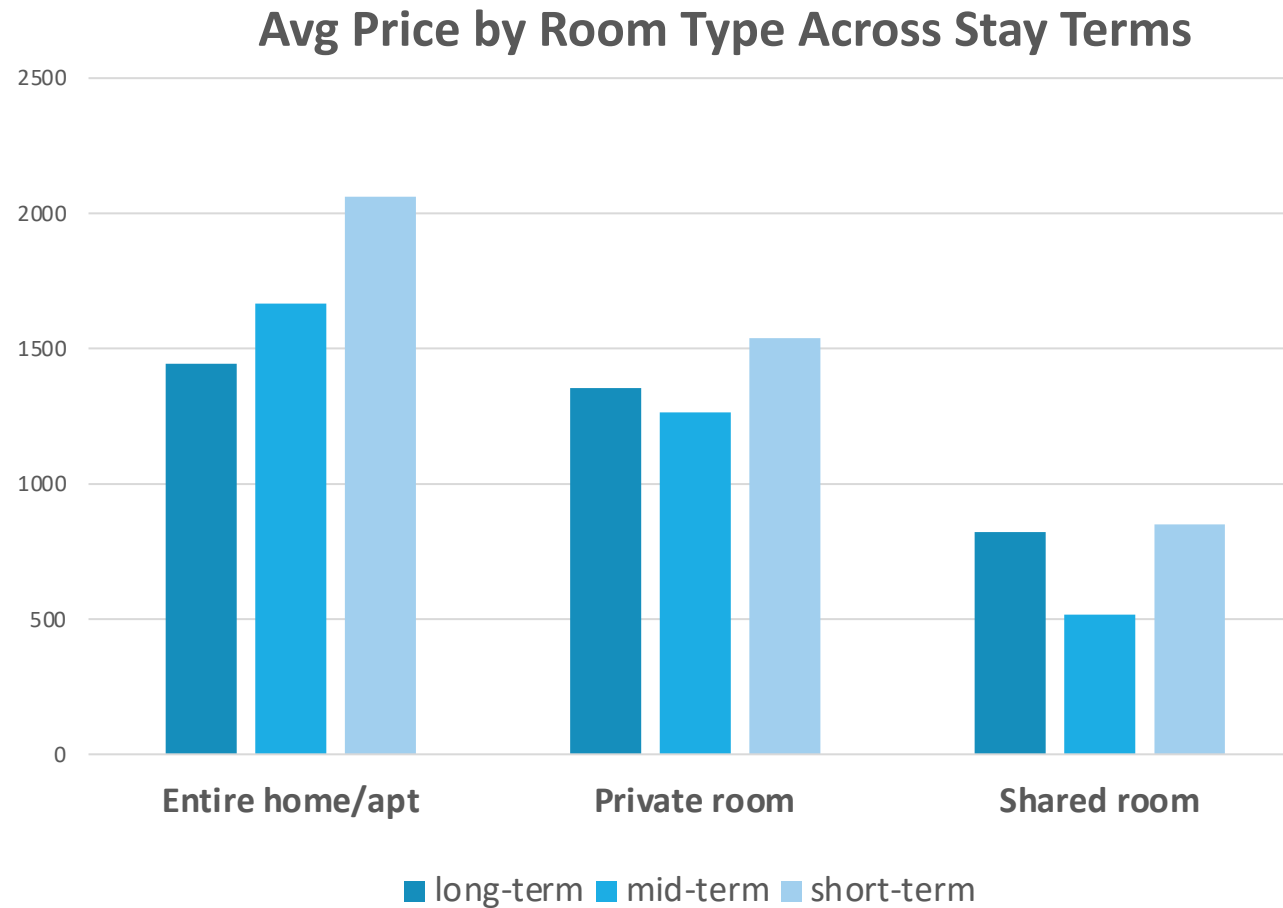
Avg Price by Room Type



Insight #4: Entire home is 20% more pricey than private room on average.

While shared room is 40% cheaper than the latter.

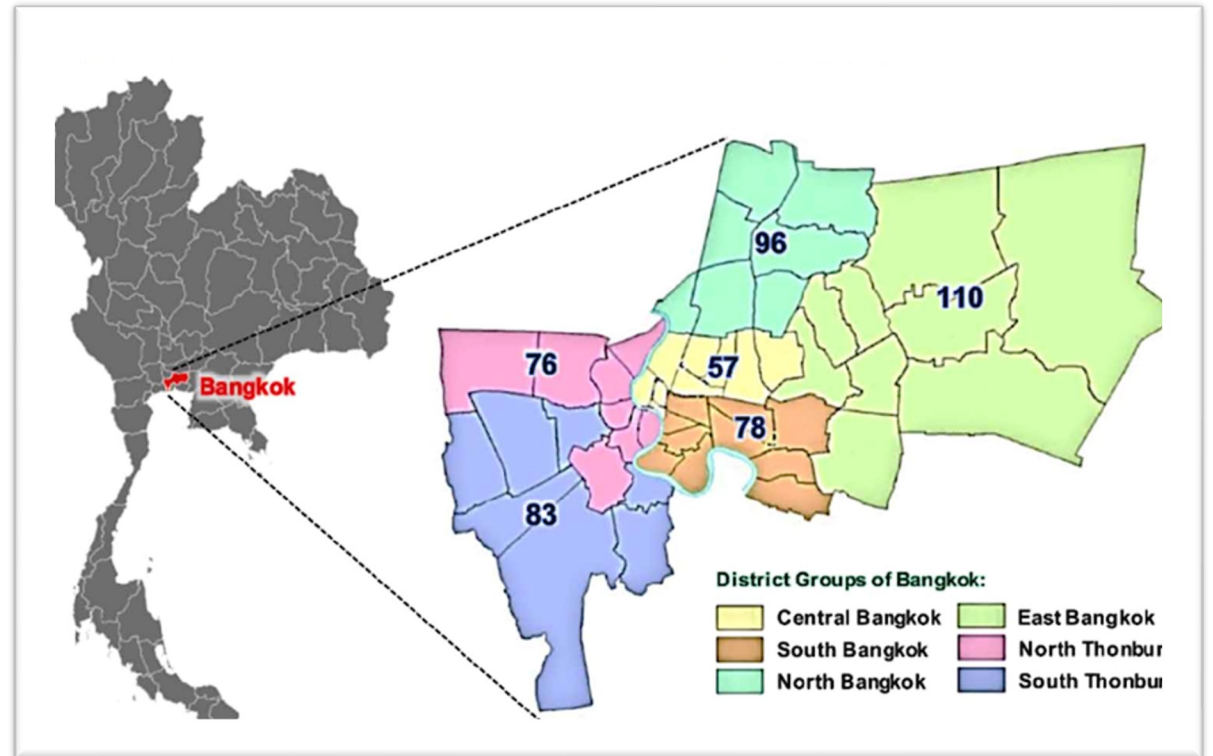
FINDINGS: ROOM TYPE



Insight #5: Short-term entire home is much pricier than private room in any stay term.

FINDINGS: LOCATION

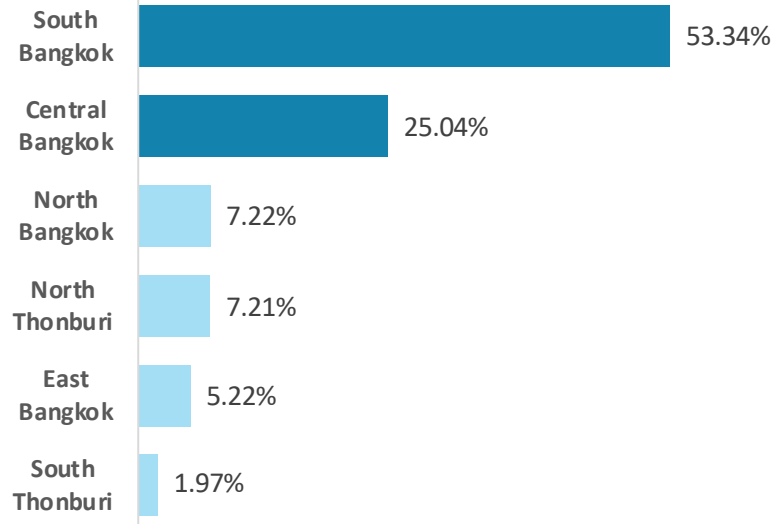
BANGKOK MAP



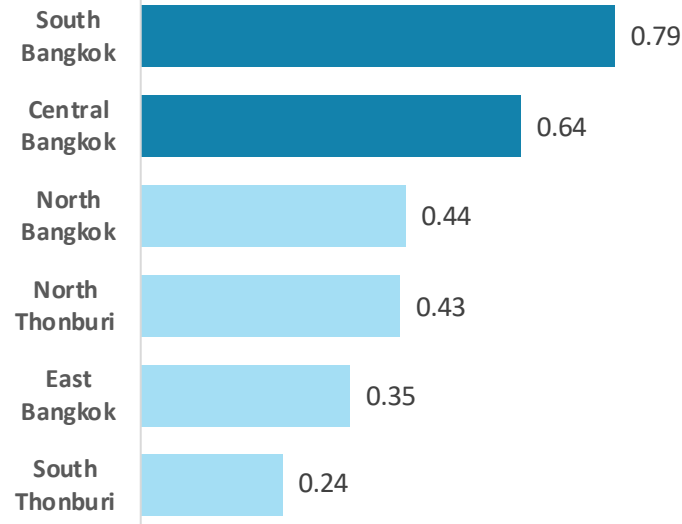
FINDINGS: LOCATION (REGIONS)

Insight #6: South Bangkok and Central Bangkok are the most popular regions in listing count, monthly reviews, and also have the highest price.

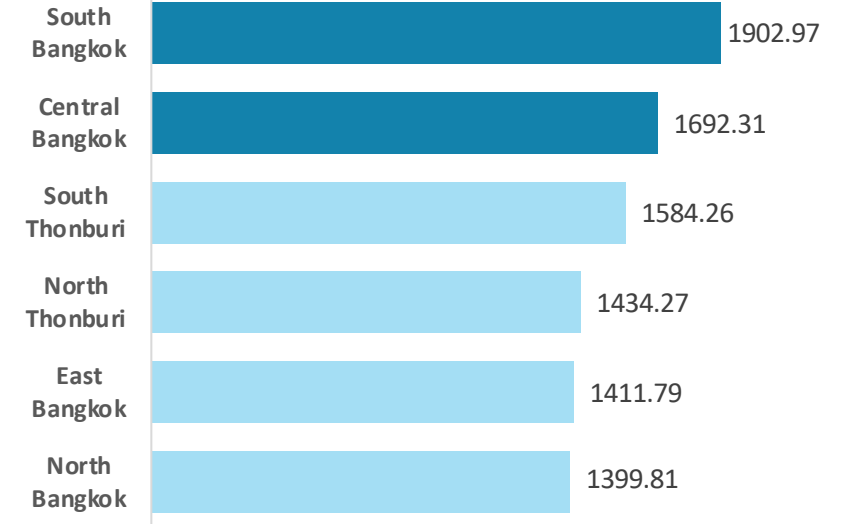
Listing Count by Region



Monthly Reviews by Region



Avg Price by Region



FINDINGS: LOCATION (REGIONS)

Insight #7: South Bangkok and Central Bangkok are also the pricier regions in for entire home and private room, but not for shared room.

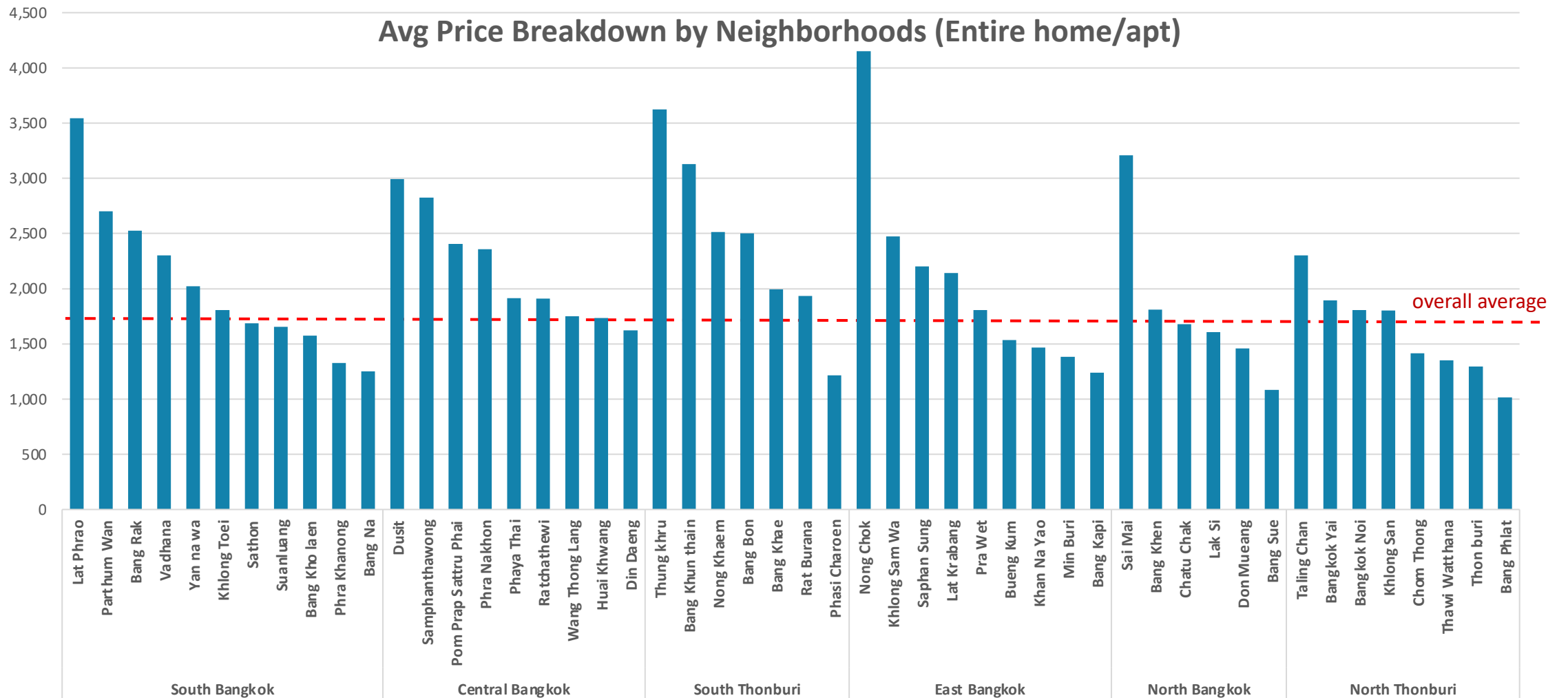
Price by Region (Entire home/apt)	
room_type	Entire home/apt
Row Labels	Average of price
South Bangkok	1964.78
Central Bangkok	1828.42
South Thonburi	1741.76
East Bangkok	1633.15
North Bangkok	1533.17
North Thonburi	1482.81
Grand Total	1861.86

Price by Region (Private room)	
room_type	Private room
Row Labels	Average of price
South Bangkok	1738.91
Central Bangkok	1419.47
North Thonburi	1383.13
South Thonburi	1382.09
North Bangkok	1260.10
East Bangkok	1156.87
Grand Total	1507.57

Price by Region (Shared room)	
room_type	Shared room
Row Labels	Average of price
East Bangkok	1288.00
North Bangkok	1005.00
North Thonburi	864.94
South Bangkok	851.39
Central Bangkok	750.68
South Thonburi	650.00
Grand Total	845.96

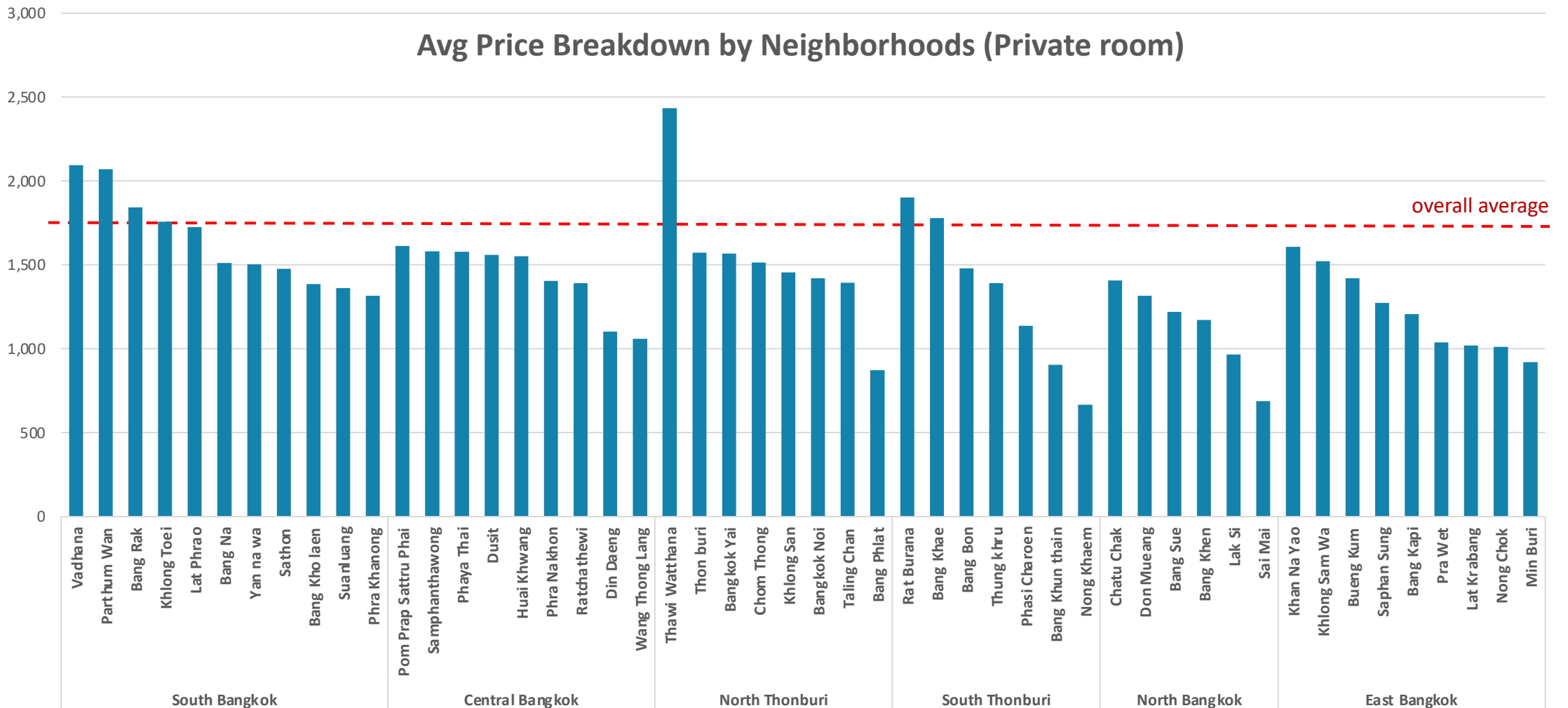
FINDINGS: LOCATION (DISTRICTS) #1

Insight #8: Price varies a lot across districts even within the same region for entire home



FINDINGS: LOCATION (DISTRICTS) #2

Insight #9: Same goes for private room but with lower level of variation.



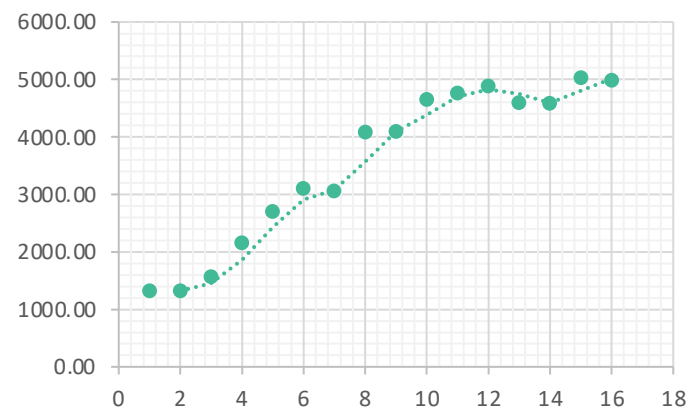
FINDINGS: CORRELATION WITH PRICE AND REVIEWS

Insight #11: Accommodates, bedrooms, and bathrooms have stronger correlation with price.

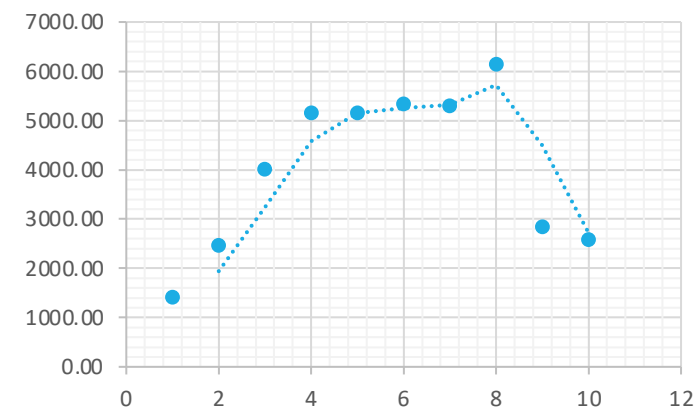
Correlation Coefficient				
	price		reviews_per_month	
accommodates	0.52		0.14	reviews_rating_score
bedrooms	0.51		0.10	accommodates
bathrooms	0.47		0.06	bedrooms
reviews_per_month	0.01		0.03	bathrooms
reviews_rating_score	-0.01		0.02	availability_365
longitude	-0.02		0.01	price
latitude	-0.03		-0.03	longitude
availability_365	-0.05		-0.05	latitude
minimum_nights	-0.14		-0.18	minimum_nights

FINDINGS: CORRELATION WITH PRICE

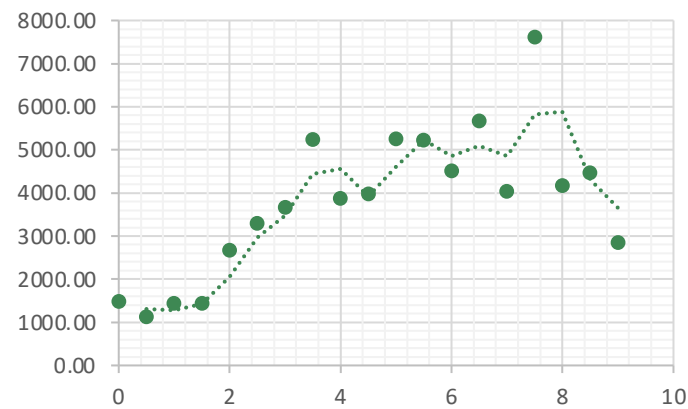
Accommodates and Price



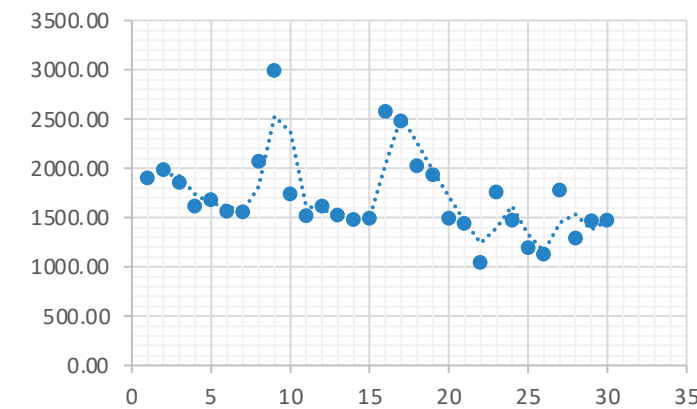
Bedrooms and Price



Bathrooms and Price

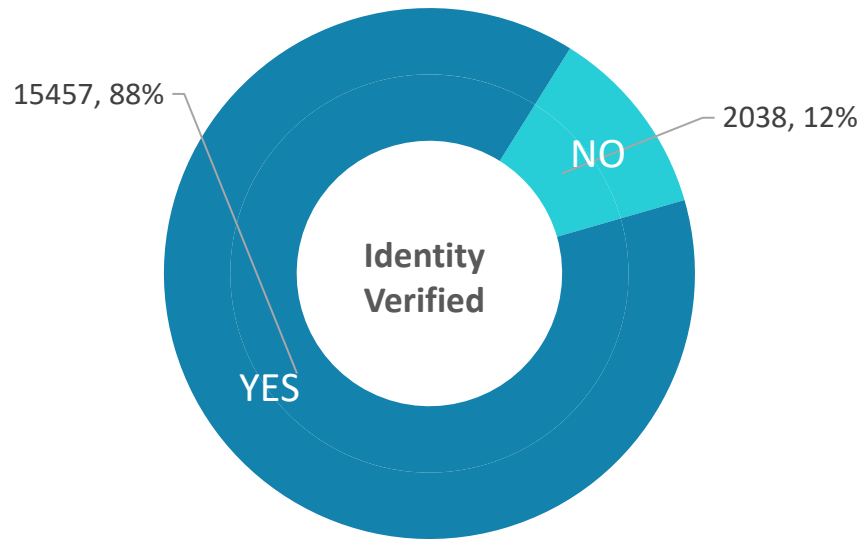


Minimum Nights and Price

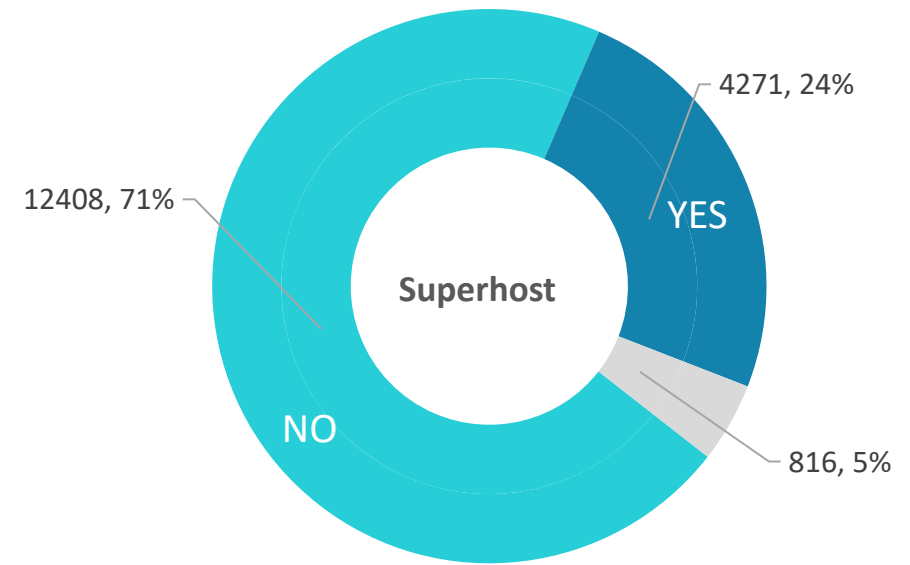


FINDINGS: EXTRA #1

Insight #12: Hosts with identity verified have more reviews, similar with superhost.



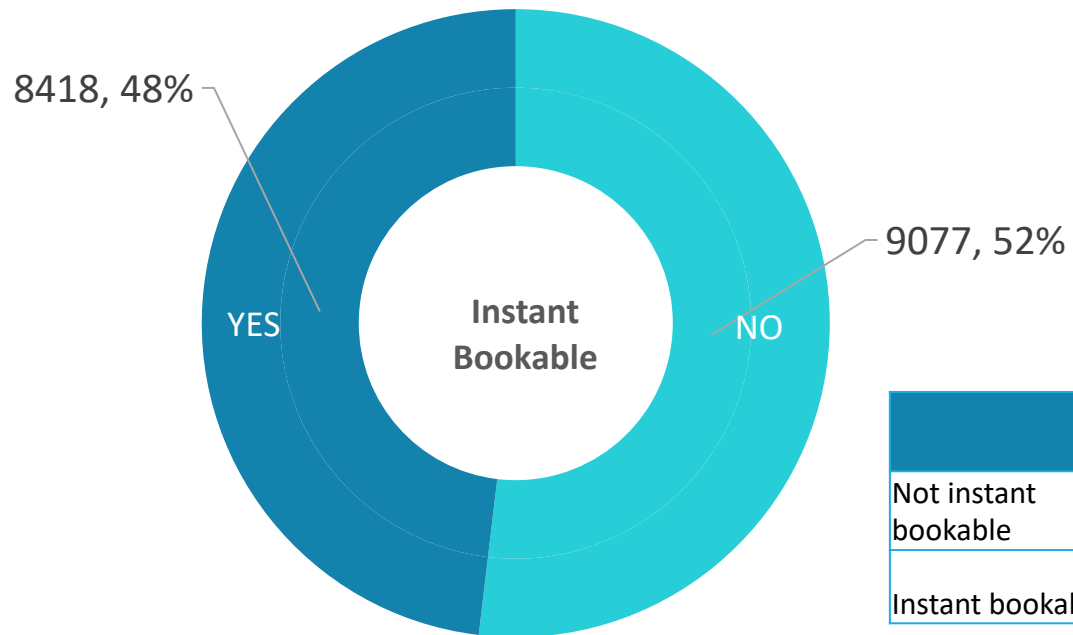
	Average price	Reviews per month
Not verified	1982.10	0.05
Verified	1717.31	0.75



	Average price	Reviews per month
Not superhost	1757.60	0.45
Superhost	1732.64	1.24
(blank)	1685.62	0.98

FINDINGS: EXTRA #2

Insight #13: Listing that is instant bookable tend to have more reviews.



	Average price	Avg reviews per month
Not instant bookable	1790.49	0.44
Instant bookable	1702.50	0.91

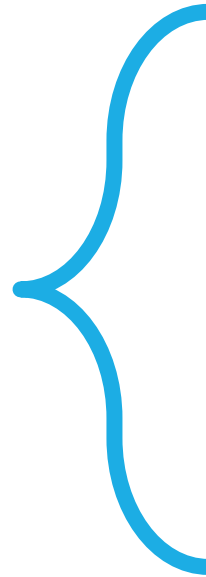
RECOMMENDATIONS

- ❑ **Room type, stay term, region, #bedrooms, #bathrooms, accommodates** are reliable factors to take into account when setting your price.
- ❑ **Allowing short-term stay** could boost your listing popularity and revenue, especially for entire home and private room.
- ❑ **Pricing slightly lower than regional average** is better to ensure revenue as there are many listings with no review having above average price across room types and regions.
- ❑ **Using districts' average price as a reference** is unreliable for the less popular districts because of smaller sample size.
- ❑ **Verify identity now and aim to be a super host** in a long run. Verified hosts have 15 times more reviews than non-verified ones.

RECOMMENDATIONS: PRICING MODEL

Suggested Price Range

(*already include 6-18% guest side service fee)



Room Type

+

Region

+

Bedrooms & Bathrooms

+

Accommodates

RECOMMENDATIONS: PRICING MODEL

“Suggested Price Range”

+

Service Fee (host-side; 3%)

+

Cleaning Fee (300-600)

+

Tax (7%)

FINAL REMARKS

- **Suggested price** range should be taken as a guideline.
- Certain **amenities** may have strong influence on price (further analysis needed).
- Other factors (outside of the dataset) such as property photos, room size, proximity to MRT or tourist landmarks, etc. could have strong correlation with price and number of bookings.